Day 4: SQL Coding Challenge – Clauses & Operators 🛒

Database: ECommerceDB

Question 1: ORDER BY & LIMIT

Scenario:

Management wants to see the top 3 highest-priced products in the e-commerce system.

Task:

- Write a SQL query to display the top 3 most expensive products.
- Sort by price in descending order.
- Use a LIMIT clause (or equivalent in SQL Server: TOP) to restrict to 3 products.

Expected Output:

Top 3 products with their product_id, product_name, and price.

Question 2: Aggregate Functions

Scenario:

Management wants summary statistics of the sales data.

Task:

- Write SQL queries using aggregate functions on the Sales table:
  - COUNT() → total number of sales records
  - SUM() → total sales amount
  - AVG() → average sale amount
  - MAX() → highest sale amount
  - MIN() → lowest sale amount

Expected Output:

Aggregated results showing total sales, average sale amount, highest and lowest sale amounts.

Question 3: GROUP BY & HAVING

Scenario:

Management wants to know which products have total sales amount greater than ₹100.

Task:

- Use GROUP BY on product_id (or product_name) to calculate total sales per product.
- Use HAVING to filter products with total sales > 100.

Expected Output:

List of products with product_id, product_name, and total sales amount greater than ₹100.

Question 4: Window Functions

Scenario:

Management wants a ranking of products based on their prices.

Task:

- Write a SQL query using a Window Function (RANK() or DENSE_RANK()) to rank products.
- Rank products from highest to lowest price.

Expected Output:

Each product with product_id, product_name, price, and rank according to price.